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Institute of Digital Technology



THE

TITANLIFT

PROJECT REPORT

Institute: Cambodia Academy of Digital Technology

Course: Algorithm & Computational Thinking 2

Lecturer: Han Leangsiv

Group: 5 Team: 9

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Presented by:

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INTRODUCTION & OVERVIEW

When people hear the word Titanic, they usually think of the famous ship. It was about 882 feet long, 92 feet wide, and built to carry up to 3,300 passengers. On its final voyage, around 2,200 people were on board. What many people don't know is that the Titanic had four elevators—three were reserved for First Class passengers, while only one was shared by everyone else.

Now, imagine bringing that same idea into a modern apartment building. Our building has 10 floors. Floor 9 is a VIP floor that can only be accessed by authorized users, while Floor 10 is an exclusive rooftop sky bar. The remaining floors are open to everyone. This inspired the idea behind our project.

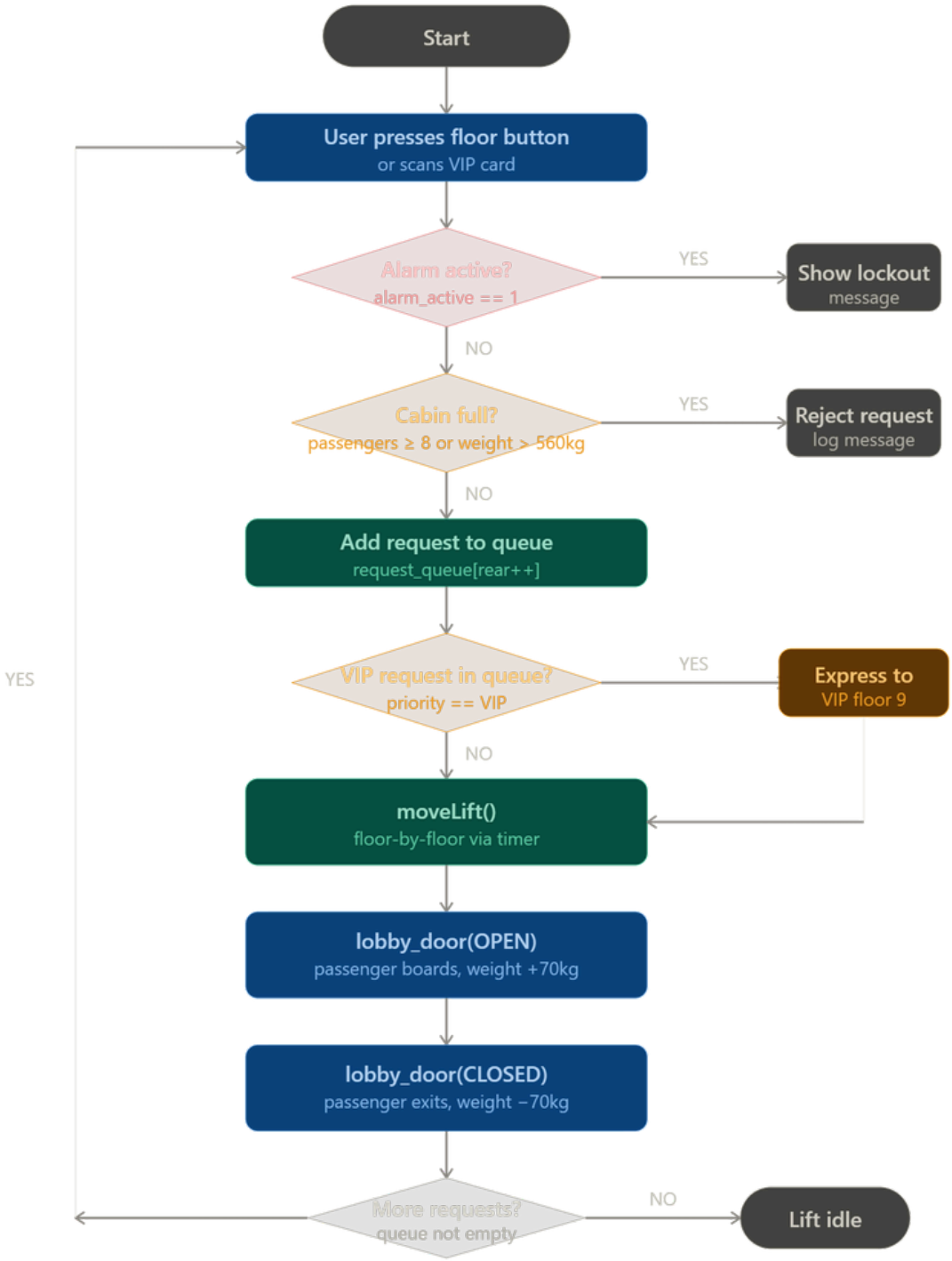
We named it TitanLift because "Titan" represents something strong, powerful, and reliable, while "Lift" simply means the elevator that takes people from one floor to another. Together, TitanLift represents a smart and powerful elevator system designed to move people efficiently.
TitanLift is a smart elevator system built in C for a 10-floor building. It keeps track of the elevator's current floor, moving direction, and request queue, while giving different priorities to normal passengers and VIP cardholders.

We built TitanLift because most traditional elevators treat every request the same, even when someone needs to get somewhere urgently. Our system gives VIP users temporary priority, but only for the moment they need it—it doesn't permanently place one group above another. That way, we can improve efficiency while still keeping the system fair for everyone.

KEY FEATURES OF SYSTEM

- ***Floor Request Queue(Priority-based):*** Stores the passenger requests, processes the highest priority first.
- ***VIP Express Mode:*** Card scan activates direct, non-stop travel to Floor 9.
- ***Card Scan Authentication:*** Validates VIP card IDs before granting Floor 9 access.
- ***Emergency Alarm Mode:*** Triggers alarm, returns the lift to Ground floor, blocks card access.
- ***Passenger Capacity Limit:*** Rejects entry once the lift reaches maximum passenger capacity when full of 8.

HOW IT WORKS


Start
User presses floor button
or scans VIP card
Alarm active?
alarm_active == 1
YES
Show lockout
message
NO
Cabin full?
passengers ≥ 8 or weight > 560kg
YES
Reject request
log message
NO
Add request to queue
request_queue[rear++]
VIP request in queue?
priority == VIP
YES
Express to
VIP floor 9
NO
moveLift()
floor-by-floor via timer
lobby_door(OPEN)
passenger boards, weight +70kg
lobby_door(CLOSED)
passenger exits, weight −70kg
More requests?
queue not empty
NO
Lift idle
YES

KEY ACHIEVEMENTS OF OUR SYSTEM

- ***VIP Express:*** When a VIP scans their card, the elevator goes straight to Floor 9 without stopping.
- ***Priority queue handles mixed request:*** It always picks the most important request first, whether it is normal or VIP.
- ***Emergency alarm respone:*** When the alarm is on, the elevator stops, goes back to the Ground floor, and opens the doors.
- ***Secure card-based access control:*** Only people with a real, registered card can use the VIP elevator. Fake or unknown cards are rejected.
- ***Passenger capacity enforcement:*** If the elevator is full of 8, it will not let more people in, because over wiegh .

CONCLUSION

- *Summary*

TitanLift started from a problem everyone's quietly felt nowsday, waiting in an elevator that stops at every floor when you're already running late. We took the Titanic's old idea of priority access and rebuilt it as something fair and temporary: a verified VIP card scan triggers Express Mode, sending the elevator straight to Floor 9, then automatically returning to normal service once that trip is done.

- *Challenges / Difficulty*

We faced many problems with the coding. We had never worked with GUI before, and our coding skills weren't strong enough yet to build the whole thing as the way we first imagined it. We had to start the GUI completely from zero, from nothing, and ended up spending almost 3 days just trying to install and configure GUI and we eventually used GTK.

CONCLUSION

- *Lesson Learnt*

During this project, we improved our coding skills in C programming. We learned how to use structures, functions, and strings together. We also learned how to solve a real problem of our own idea instead of only doing textbook exercises. And this was our first time creating a GUI, so we gained new knowledge and experience. Most important, we learned how to work well as a team and help each other to finish the project.

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THANK YOU

For your time and attention

References

Claude AI (Anthropic, 2026). Available at: claude.ai

Concept inspired by: The Leaf Coffee
(Noden2 Hotel & Apartment)

Naming & priority concept inspired by: RMS Titanic (1912)
— reimagined as a fair, temporary access system

The TitanLift

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